

ANJALI SAINI

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PROFESSIONAL SUMMARY

M.Sc. Bioinformatics candidate with expertise in Python, molecular docking, virtual screening, NGS, Galaxy workflows, functional annotation, pharmacogenomics, biological data interpretation, and machine learning. Experienced in in-silico drug discovery, protein-ligand interaction analysis, bioinformatics pipelines, and various bioinformatics tools. Proven ability to process large-scale datasets (50,000+ records) and develop predictive models with >85% accuracy. Proactive and quick learner with strong communication skills and a growth mindset, seeking to contribute to innovative bioinformatics and genomics projects in precision medicine.

EDUCATION

Master of Science — Bioinformatics

Maharshi Dayanand University, Rohtak, Haryana

Aug 2024 — June 2026

Bachelor of Science — Biotechnology

Dronacharya Govt. College, Gurugram | First Division

July 2021 — July 2024

WORK EXPERIENCE

Computational Biology Intern — Growdea Technologies Pvt. Ltd.

Dec 2025 — Jan 2026

- Processed and structured 50,000+ pharmaceutical records using Python (pandas) and regex-based text mining, improving data readiness by ~40%.
- Managed 3+ in-silico pipelines, enhancing data processing efficiency by 30%..

DISSERTATION RESEARCH

In Silico Evaluation of Phytochemicals Against Antibiotic-Resistant *Klebsiella pneumonia* | M.Sc. Dissertation | 2025–2026

- Performed molecular docking of 102 phytochemicals (filtered from 36,464 compounds) against PBP targets using AutoDock and Raccoon; identified top 25 candidates based on binding affinity.
- Prepared and validated protein structures from PDB and AlphaFold; evaluated drug-likeness using ADMET and Lipinski's Rule of 5, reducing candidates by ~70%.
- Conducted structural and comparative analysis using UCSF Chimera and PyMOL; generated reports with docking visualizations for research interpretation.

Applied protein structure validation, active-site identification, interaction analysis, and ADMET profiling using AlphaFold2, UCSF Chimera, AutoDock 4.2, RACCOON, LigPlot+, and Discovery Studio.

ACADEMIC PROJECTS

Whole Genome Sequencing and Comparative Antimicrobial Resistance Analysis of Carbapenem-Resistant *Klebsiella pneumoniae* using Galaxy

- Performed whole genome sequencing and resistome analysis of isolates from India, Turkey, and the United States using Galaxy Europe. tools including FastQC, Trimmomatic, BWA-MEM2, FreeBayes, and ABRicate. Identified 30,261 genomic variants and region-specific carbapenemase genes such as blaNDM-1, blaKPC-2, and blaKPC-3.

Hospital Overview Dashboard | Power BI

- Built a 3-page interactive Power BI dashboard on a healthcare dataset, analyzing 200+ appointments and ₹551K revenue; enabled faster decision-making through KPI visualization and trend analysis.

Machine Learning Pipeline — Biomedical Text Analytics | Orange Data Mining

- Developed end-to-end machine learning workflows for biomedical text preprocessing, classification, and interpretation; achieved 85%+ accuracy and improved model performance through optimized feature selection.

Machine Learning on Diabetes Dataset | WEKA

- Implemented classification algorithms (Naive Bayes, Decision Tree, SVM) on a 768-record diabetes dataset; performed EDA and feature engineering, improving prediction accuracy by 12%.

Drug Discovery & Molecular Docking | HEX

- Designed molecular docking workflow using HEX; analyzed protein–ligand interactions and prepared chemical structures using UCSF Chimera and ChemSketch.

Python Programming Projects | GitHub Portfolio

- Developed multiple command-line Python applications, including Blackjack, Hangman, Calculator, Secret Auction, and Password Generator. Applied core programming concepts such as functions, loops, conditional statements, dictionaries, and modular code organization

CERTIFICATIONS & ACADEMIC ACTIVITIES

Academic Poster Presentation on NGS Workflows, Maharshi Dayanand University, Rohtak

Introduction to Cloud Computing — Online Certification

AI & Machine Learning in Pharmaceutical Sciences — Maharshi Dayanand University

Hands-on Workshop on Bioinformatics & Molecular Docking BioSoc-DTU

Tata Group Data Analytics Job Simulation — Forage

Deloitte Australia Data Analytics Job Simulation — Forage

McKinsey Forward Program — McKinsey & Company

SKILLS

Technical Skills: Molecular Docking, Virtual Screening, Protein–Ligand Interaction, ADMET, BLAST, NGS Workflows, Scikit-learn, pandas, Feature Engineering, Statistical Analysis, Python, SQL, PostgreSQL, PyMOL, UCSF Chimera, AutoDock, AlphaFold, Power BI, Bioinformatics tools

Soft Skills: Analytical Thinking, Problem-Solving, Research Skills, Attention to Detail, Team Collaboration, Time Management

Research & Professional Skills Literature Review, Scientific Skills, Workflow Development, Presentation Skills, Biological Data Analysis, Data Visualization, NGS Workflow, Data Annotation, Functional Annotation

LANGUAGES

English (Professional) | Hindi (Native)